3 **(a)** Using $pV = nRT$,

$$n = \frac{pV}{RT} = \frac{1.0 \times 10^5 \times 6.0 \times 10^{-4}}{8.31 \times 300}$$
$$= 0.024067 = 0.024 \text{ mol}$$

(b) Using the ideal gas equation $pV = nRT \Rightarrow n = \frac{pV}{RT}$.

For a fixed mass of gas, n is the same at both the initial state (1) and final state (2).

$$\therefore \frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$

$$p_2 = \frac{p_1V_1T_2}{T_1V_2} = \frac{1.0 \times 10^5 \times 6.0 \times 10^{-4} \times 550}{300 \times 5.0 \times 10^{-5}}$$
$$= 2.2000 \times 10^6 = 2.2 \times 10^6 \text{ Pa}$$

Alternatively using state (1),

$$p_2 = \frac{nRT_2}{V_2} = \frac{0.024067 \times 8.31 \times 550}{5.0 \times 10^{-5}}$$
$$= 2.19996 \times 10^6 = 2.2 \times 10^6 \text{ Pa}$$

(c) Since the gas can be treated as an ideal gas, increase in internal energy ΔU is given by

$$\Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} \times 0.024 \times 8.31 \times (550 - 300)$$
$$= 74.79 = 75 \text{ J}$$

Alternatively,

$$\Delta U = \frac{3}{2} \Delta(pV) = \frac{3}{2} \times \left[(p_2V_2 - p_1V_1) \right]$$
$$= \frac{3}{2} \times \left[\left(2.2 \times 10^6 \times 5.0 \times 10^{-5}\right) - \left(1.0 \times 10^5 \times 6.0 \times 10^{-4}\right) \right]$$
$$= 75 \text{ J}$$

(d) **(i)** The first law of thermodynamics states that the increase in the internal energy of a system is equal to the sum of the heat supplied to the system and the work done on the system, and the internal energy of a system depends only on its state.

(ii) The gas loses heat hence Q is negative. Compression stroke is rapid to minimise heat loss from the gas to the engine block/surroundings, so that Q is less negative.

Since gas is compressed, work done on it W is positive. From first law of thermodynamics $\Delta U = W + Q$. If Q is less negative, the gain in internal ΔU is maximised.

For ideal gas, $\Delta U \propto \Delta T$ where ΔT is the temperature increase during the compression stroke and ΔT will correspondingly be maximised. This leads to higher efficiency for the engine.
OR

When the compression stroke takes place rapidly, heat loss from the gas to the surroundings is minimised such that Q can be assumed to be 0.

Since gas is compressed, work done on it W is positive. From first law of thermodynamics $\Delta U = W + Q$. Since $Q = 0$ (and not negative i.e. there is heat loss from the gas), the gain in internal ΔU is maximised.

For ideal gas, $\Delta U \propto \Delta T$ where ΔT is the temperature increase during the compression stroke and ΔT will correspondingly be maximised. This leads to higher efficiency for the engine.